

OMER GADALLA

Aerospace Engineer

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EDUCATION

Middle East Technical University (METU)

2021–2026

B.Sc. Aerospace Engineering

EXPERIENCE

Galaxy Rocket Team

December 2024–Present

Aerodynamics Engineer

TEKNOFEST Liquid Bipropellant Rocket Competition (A7) | 2026

- Contribute to system-level design of a liquid bipropellant rocket to meet competition targets for total impulse, apogee, and propellant mass fraction.
- Conduct external aerodynamic CFD analyses to quantify drag coefficients and support flight performance.
- Perform liquid bipropellant engine performance analyses using CFD and FEA to validate engine design decisions.

Propulsion Analysis Engineer

TEKNOFEST Liquid Bipropellant Rocket Engine Development Competition (B3) | 2025

Design & Analysis Phase

- Contributed to the design and analysis of a 100N liquid bipropellant rocket engine across feed system, combustion, and nozzle subsystems.
- Developed a propellant feed system model in Simscape to analyze engine's dynamic behavior and system interactions.
- Performed combustion and nozzle CFD analyses to investigate flow behavior and evaluate engine performance.
- Conducted thermo-structural FEA to assess structural integrity under high pressure and high temperature.

Testing Phase

- Participated in spray injector cold-flow testing to evaluate atomization characteristics and correlate experimental results with CFD simulations.
- Executed multiphase DPM CFD to analyze injector atomization characteristics and finalized results for the design team.
- Assisted in instrumentation and control components testing, including functional, performance, and safety verification.
- Achieved Finalist ranking among national teams.

CFD & Structural Analysis Team Lead

TEKNOFEST High Altitude Rocket Competition (A3) – 2025

- Led CFD and structural analysis efforts for a high-altitude rocket designed to reach Mach 1.4 and 15,000 ft apogee.
- Developed external aerodynamic CFD simulations to quantify drag and support flight performance evaluation.
- Performed FEA structural analysis to ensure integrity under high-altitude flight conditions.

MTS Defense Industry

July 2025–September 2025

R&D Engineer – Intern

- Developed the conceptual design of a 20 N green bipropellant thruster for small satellite propulsion applications.
- Designed and modeled the propellant feed system to assess system behavior and integration feasibility.
- Presented the project to investors, communicating system architecture, technical maturity, and key development risks.

Skills

CAD: Inventor, SolidWorks

|| Structural: Ansys, Nastran

|| Programming: MATLAB, Simulink, Simscape

CFD: Ansys FLUENT, OpenFoam

|| Languages: English (Fluent), Turkish (Fluent), Arabic (Fluent)

Portfolio

For a summary of my engineering projects I have worked on and my skills, check out my online portfolio at:

<https://omergadalla.github.io/portfolio/>